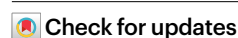


Navigating energy transition solutions for climate targets with minerals constraint

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Urgent decarbonization is imperative, yet mineral scarcity may threaten low-carbon technology deployment, potentially challenging transition pathways. Here, through the analysis of 557 mitigation pathways from the Intergovernmental Panel on Climate Change Sixth Assessment Report using the Global Resource Evaluation of Abatement Technologies model, we systematically quantify demand and potential shortages for 40 minerals critical to 17 energy technologies. We find that all pathways may experience global shortages of up to 12 minerals by 2100 under the moderate scenario, with more severe shortages of indium, tin, cadmium and tellurium related to thin-film photovoltaic, wind and nuclear power (>50% of pathways). Regional disparities would intensify these risks, particularly in developing, resource-vulnerable regions (for example, the Middle East and Africa), with potential shortages reaching 24 minerals. Hence, we propose better solutions that balance emissions reduction with resource constraints and regional equity, revealing that decarbonization requires more than technological innovation. It demands the strategic integration of diversified energy technology portfolios, aggressive recycling, material substitution and global trade cooperation, alongside moderate gross domestic product growth.

Global warming is accelerating at an unprecedented pace, with the 2024 global average temperature rise reaching approximately 1.55 °C above pre-industrial levels¹—surpassing the Paris Agreement’s critical 1.5 °C threshold and endangering critical ecosystems and human livelihoods. The immediate decarbonization of energy systems is imperative^{2,3}. However, identifying viable transition pathways within this narrowing time window presents great challenges, requiring not only fundamental socio-economic transformations but also sustainable access to critical resources for manufacturing low-carbon technologies⁴, such

as copper, dysprosium, neodymium, indium and lithium^{5–11}. While scientific models have proposed numerous mitigation pathways for 1.5 °C and 2 °C targets, the complex mineral resource constraints and affordability concerns may largely threaten their feasibility. However, the resource–climate nexus across transition pathways is currently not well understood, which results in policymakers facing the daunting challenge of navigating hundreds of seemingly ‘perfect’ pathways. A careful balance between climate action and potential mineral resource risk is therefore crucial.

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Scholars have conducted plenty of research to evaluate the mineral demand and identify the likely shortages, providing valuable insights for understanding the resource–climate nexus^{12–14}. However, they either focussed on limited technological subsets (for example, wind and photovoltaic (PV) power^{15–17}), restricted transition pathways (for example, International Energy Agency scenarios)^{13,18–20} or a narrow range of critical minerals^{6,8,11}. These fragmented analyses may generate potentially biased risk assessments and incomplete transition roadmaps²¹. More critically, it fails to answer how to best balance climate ambitions with resource constraints. Consequently, a systematic analysis encompassing the full spectrum of socio-economic–technological transition options and critical minerals is urgently required to develop feasible solutions that can simultaneously address climate mitigation and resource security concerns.

This study advances the field by developing the Global Resource Evaluation of Abatement Technologies model, which systematically quantifies global and regional demand and shortage risks for 40 critical minerals across 17 technologies under all pathways in the Sixth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC AR6), while proposing better 1.5 °C and 2 °C transition solutions through synthesizing global mineral constraints and regional disparities across temporal scales (Supplementary Figs. 1 and 2 and Methods). This study encompasses the most extensive set of mitigation pathways to date (557 pathways in total: 367 for 2 °C and 190 for 1.5 °C)²² and incorporates a complete spectrum of supply-side energy technologies and their associated critical minerals. The proposed quantitative solution integrates a dual-tier approach: (1) emissions reduction strategies incorporating optimized socio-economic paradigms, technological portfolios and energy structures with minimum mineral resource risks; and (2) comprehensive mineral-focus strategies spanning enhanced regional trade networks and accelerated technological innovation. Our research provides insights into mineral demand, illuminating the challenges that lie ahead if current proposed pathways are followed. It can navigate policymakers towards better actions in terms of safety, feasibility and equity.

Global critical mineral shortages impeding climate targets

We first evaluate the mineral shortage risks associated with 557 IPCC AR6 pathways for the 2 °C and 1.5 °C targets. Considering that the future mineral demand strongly depends on the mineral intensity and recycling efficiency, this study develops 80 scenarios to represent such uncertainties, and on this basis systematically visualizes the mineral shortage risk landscape (Fig. 1, Methods and Supplementary Data).

Our panoramic analysis reveals that even aggressive improvements in mineral intensity and recycling efficiency may still face shortage risks. Projections to 2100 indicate that, compared with the proven economic reserves of 2021, up to 13 minerals may face shortages under 2 °C pathways if progress in intensity reduction and recycling efficiency remains conservative (Fig. 1a). However, this number could decrease to 11 if intensity reaches the R3 level. Achieving the R1 level for intensity, along with a rapid increase in recycling rates, could prevent shortages for up to seven minerals (Fig. 1b,c). Considering both technical feasibility and length restrictions, subsequent analysis will primarily focus on moderate scenarios unless otherwise specified, with results of aggressive and conservation scenarios provided in Supplementary Figs. 6–11.

Under the moderate scenario, our analysis reveals that all 557 IPCC AR6 pathways would suffer potential shortages, ranging from 1 to 12 critical minerals (In, Ir, Cd, Sn, Te, Ag, Ge, Tb, Zn, C, Li and Ni), based on projected mineral demands under current technological development trends. On average, the 2 °C and 1.5 °C pathways exhibit shortages of five and seven minerals, respectively—most frequently indium, cadmium, tellurium and tin, which are essential for thin-film solar PV, wind and nuclear power (Fig. 2 and Supplementary Fig. 2). The average cumulative consumption of minerals subject to potential shortage under 2 °C pathways ranges from 0.3 kt to 133.2 Mt (Fig. 2), with indium

experiencing the most severe shortages (demand exceeding reserves by 8.2-fold across >80% of pathways). The gaps are substantially wider under the 1.5 °C scenarios. Temporal projections in the 2 °C pathways indicate global indium demand surpassing available reserves by 2035 on average, followed by cadmium (2060), iridium (2065), tin (2085) and tellurium (2100). Such shortages could undermine the feasibility of current technological development pathways, underscoring the importance of timely adjustments in technological alternatives and resource exploration. Taking solar PV as a typical example, the shortage of some constraining minerals associated with thin-film technologies (for example, indium, cadmium and tellurium) could be alleviated through high-efficiency crystalline silicon alternatives or breakthroughs in mineral-reduced thin-film solutions; conversely, if current technological trajectories persist, accelerated exploration could be essential, particularly for indium (Supplementary Fig. 4). To this end, we outline a cautionary timeline for critical mineral exploration breakthroughs on the basis of the global average lead time from discovery to production (approximately 15.7 years)²³. Under the 1.5 °C pathways, these shortages could occur 5–25 years earlier, with peak shortages reaching 11.2 times current reserve levels for certain minerals.

Because the projected demand surge may render previously uneconomical reserves viable, we also compare the mineral demands with the reserve base. Results show that, even with the optimistic progress where all currently proven reserve base become economically extractable, there are still potential shortages of up to 11 minerals across both the 2 °C and 1.5 °C pathways by 2100, with only lithium and germanium probably achieving sufficiency in this specific case (Supplementary Fig. 5). This critical gap underscores the imperative for accelerated exploration and new geologic discoveries of critical mineral resources. If mineral exploration mirrors the progress observed in petroleum development (where proven oil reserves increased 1.5-fold from 1980 to 2020)²⁴, shortages of Ag, C, Cd, Ge, Li, Ni, Sn, Tb, Te and Zn could be avoided across all 2 °C pathways, with similar benefits for most minerals (all except Cd) under the 1.5 °C pathways.

It should be noted that the identified shortages may be reshaped by material substitutions (for example, synthetic graphite) and emerging technologies such as cobalt-free batteries (for example, lithium iron phosphate and lithium nickel manganese oxide) or next-generation thin-film PV with reduced critical material requirements.

Regional disparities heighten mitigation challenges and inequities

Considering that regional disparities in resources may intensify these risks, we extend the analysis to a regional resolution across all ten IPCC AR6 regions, with projections following current technological development trends.

Our vulnerability assessment reveals that 24 minerals would exhibit regionally varying degrees of inadequacy across all 2 °C pathways (Fig. 3a and Methods). Some less-developed regions are projected to suffer more severe mineral shortages and heightened vulnerability risks. REST_ASIA emerges as the most vulnerable with nine critical minerals (Mo, B, Cr, Mn, V, Nb, Hf, Ta and Ge) exceeding shortage thresholds. MIDDLE_EAST and INDIA+ follow closely, each confronting acute shortages of seven minerals (including B, V, Nb, Ta and so on). AFRICA and EU28+ may experience substantial scarcity of five critical minerals (including Nb, Ta, Ge and so on). In contrast, other regions demonstrate substantially lower vulnerability, with only two–four minerals surpassing shortage ratios of 10 across all pathways.

Region-specific mineral shortages follow distinct geographical patterns. Indium deficits are particularly critical, with all ten regions facing supply constraints. Nine regions need additional tellurium (all except LATIN_AM) and iridium (all except AFRICA), while eight regions may face deficits in tantalum, lithium, tin and zinc (all except LATIN_AM and PACIFIC_OECD), as well as niobium (all except LATIN_AM and NORTH_AM) and tellurium (all except CHINA+ and PACIFIC_OECD).

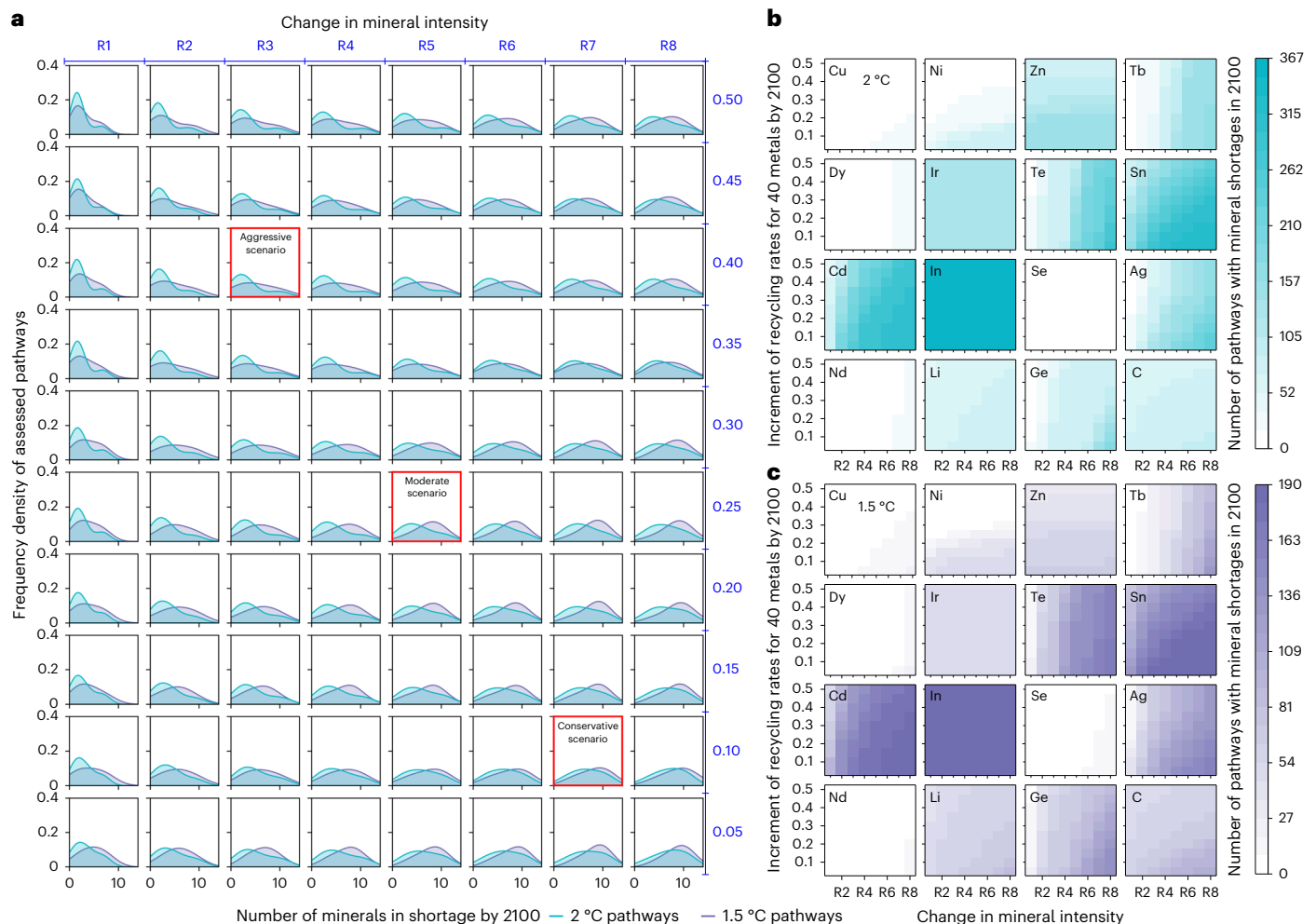


Fig. 1 | Global mineral shortage projections under 80 scenarios by 2100.

a, The frequency distribution of the number of minerals experiencing shortages across all the 1.5 °C (purple) and 2 °C (blue) pathways, with each box representing a single scenario (80 in total). **b**, The probability of shortage of individual minerals across 367 pathways for the 2 °C target. **c**, The probability of shortage of individual minerals across 190 pathways for the 1.5 °C target. The 80 scenarios are designed by combining different levels of recycling efficiency and mineral intensity. Specifically, since baseline recycling rates vary across technologies, we set ten levels for the increment of recycling rates by 2100, with a linear progression relative to baseline values, ranging from 0.05 to 0.5. Here, <0.20 indicates slow improvements, 0.20–0.40 indicates medium improvements and

>0.40 indicates rapid improvements. Additionally, eight levels (R1–R8) are set for mineral intensity reduction, each indicating a decline to the most advanced level reported in the literature by a specified year, followed by a continued decline at half the preceding rate until 2100 (see the Methods for details). Specifically, R1–R8 correspond to reaching the most advanced level in 2030, 2040, 2050, 2060, 2070, 2080, 2090 and 2100, respectively. R1 represents the case with the most rapid technology innovation, while R8 is the case with the slowest progress. Both baseline and advanced values are derived from the established literature (Methods, Supplementary Figs. 15–17 and Supplementary Data 5–6), with three representative scenarios (red boxes in **a**) analysed in detail. The projections to 2050 are presented in Supplementary Fig. 3.

Seven regions grapple with shortages of tin (all except LATIN_AM, REST_ASIA and PACIFIC_OECD), platinum (all except LATIN_AM, RUSSIA+ and PACIFIC_OECD), molybdenum (all except CHINA+, LATIN_AM and NORTH_AM) and germanium (all except CHINA+, NORTH_AM and RUSSIA+). Beyond these widespread shortages, RUSSIA+ additionally faces deficits in hafnium and silver, while CHINA+ confronts further shortages of chromium, copper and hafnium. When assessed by the number of minerals in shortage, developing regions demonstrate particularly acute additional vulnerabilities: MIDDLE_EAST (11 extra mineral deficits including hafnium, chromium, vanadium, selenium and manganese), REST_ASIA (9 additional minerals), INDIA+ (8 additional minerals) and AFRICA (6 additional minerals). In contrast, some developed regions experience fewer but still substantial additional shortages, with NORTH_AM lacking four extra minerals (nickel, chromium, manganese and vanadium), while PACIFIC_OECD faces two additional deficits (boron and chromium). These shortages hinder the deployment of region-specific technologies. For example, 48% of

the EU28+’s expected solar PV capacity may be delayed owing to tin shortages starting in 2070, while the Middle East, facing shortages earlier since 2050, could see 85% of its anticipated capacity constrained.

This disparity would be amplified for reaching more ambitious climate targets. Under the 1.5 °C pathways, shortages are projected to intensify across 16 critical minerals (Co, Mg, Ga, Ge, Te, Se, Cd, Al, Ag, Li, C, W, In, Ta, Ni and Cu). While the PACIFIC_OECD experiences relatively moderate deficit increases of 1–10% higher deficits compared with 2 °C pathways, the other nine regions face substantially higher shortages, with increases ranging from 5% to 42% (Fig. 3b). Notably, MIDDLE_EAST’s vulnerable mineral portfolio (with shortage ratios >10 across all assessed pathways) expands to eight types, while other regions maintain their 2 °C levels.

Optimal and balanced solutions

According to the above global and regional risk analysis, all the assessed pathways show substantial disparity in terms of mineral

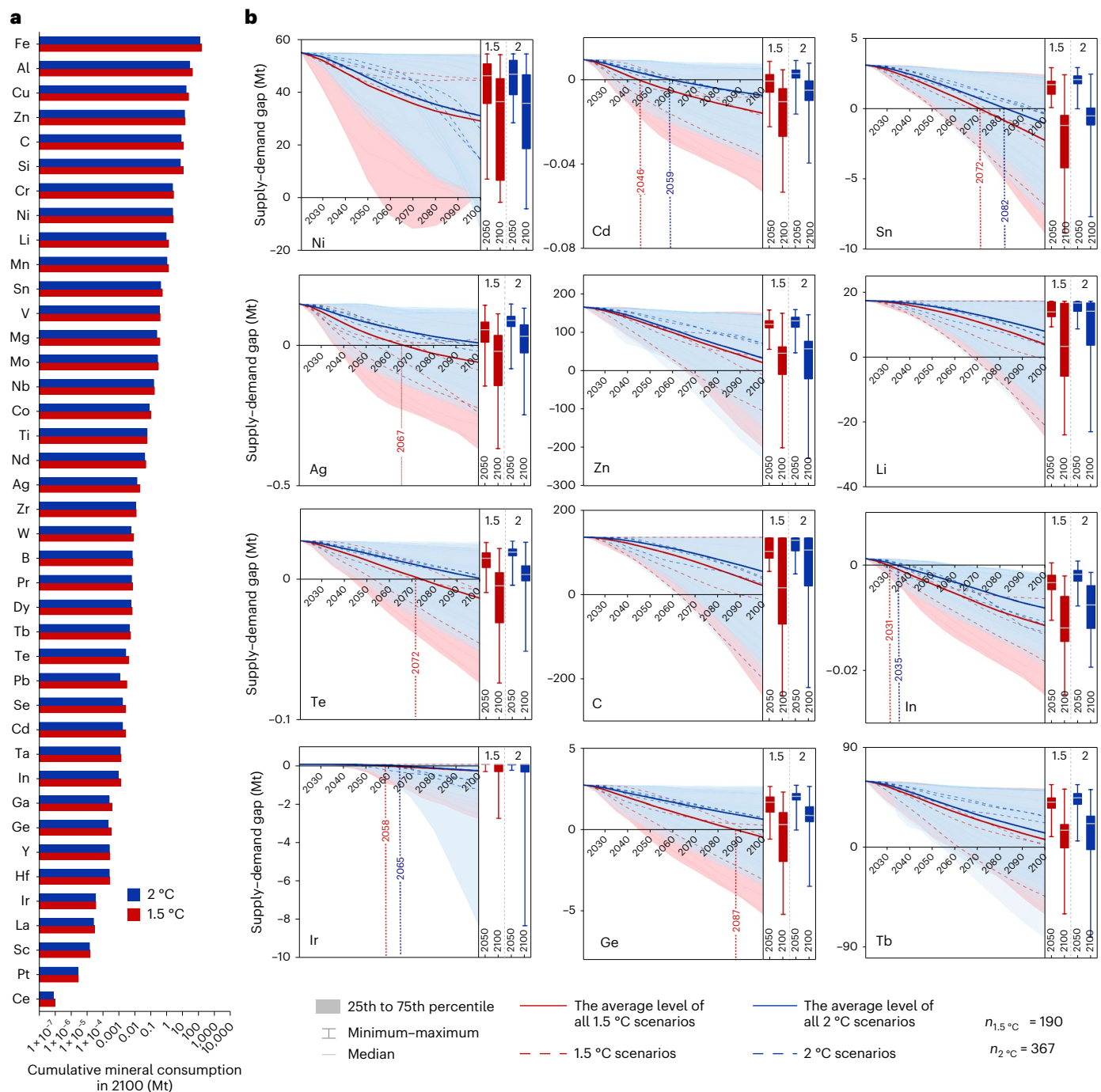


Fig. 2 | Projected demand and shortages of critical minerals across the 1.5 °C and 2 °C pathways under the moderate scenario (2005–2100). **a**, The cumulative consumption of 40 minerals in 2100. **b**, The annual supply–demand gap of 12 minerals under 557 mitigation pathways. The result here is for the moderate innovation scenario. The vertical dashed line in **b** represents the projected onset years of average-level mineral shortages in mitigation pathways (red for 1.5 °C and blue for 2 °C). Note that the cumulative mineral consumptions in **a** are shown using a logarithmic scale. On average, five minerals exhibit

shortages in the 2 °C pathways (indium, cadmium, tellurium, tin and iridium), and seven in the 1.5 °C pathways (indium, cadmium, tellurium, tin, silver, germanium and iridium). A total of 34% of 2 °C and 20% of 1.5 °C pathways show shortages of three or fewer minerals, most commonly indium, cadmium and tin, while 18% and 19% of 2 °C and 1.5 °C pathways, respectively, involve shortages of nine or more minerals. The results under aggressive and conservative scenarios are presented in Supplementary Figs. 6 and 7.

shortages, degree of shortage, times of shortage and coverage regions. Therefore, how to navigate the transition pathway within the constrained time window to simultaneously achieve effective emission reduction and ensure mineral resource security becomes very complicated. Aiming to answer this challenging but important question, we first establish two indices, the Global Mineral Shortage Index (GMSI) and Regional Mineral Shortage Index (RMSI), to

identify the optimal emissions reduction strategies with minimum mineral risks across all IPCC AR6 pathways aligned with the 1.5 °C and 2 °C targets (Methods). These two indices can systematically assess how mitigation pace, socio-economic shifts and technological portfolios influence mineral risks across pathways. Following that, we propose comprehensive mineral-focus strategies to solve the global and regional mineral shortages that remain in the optimal pathways,

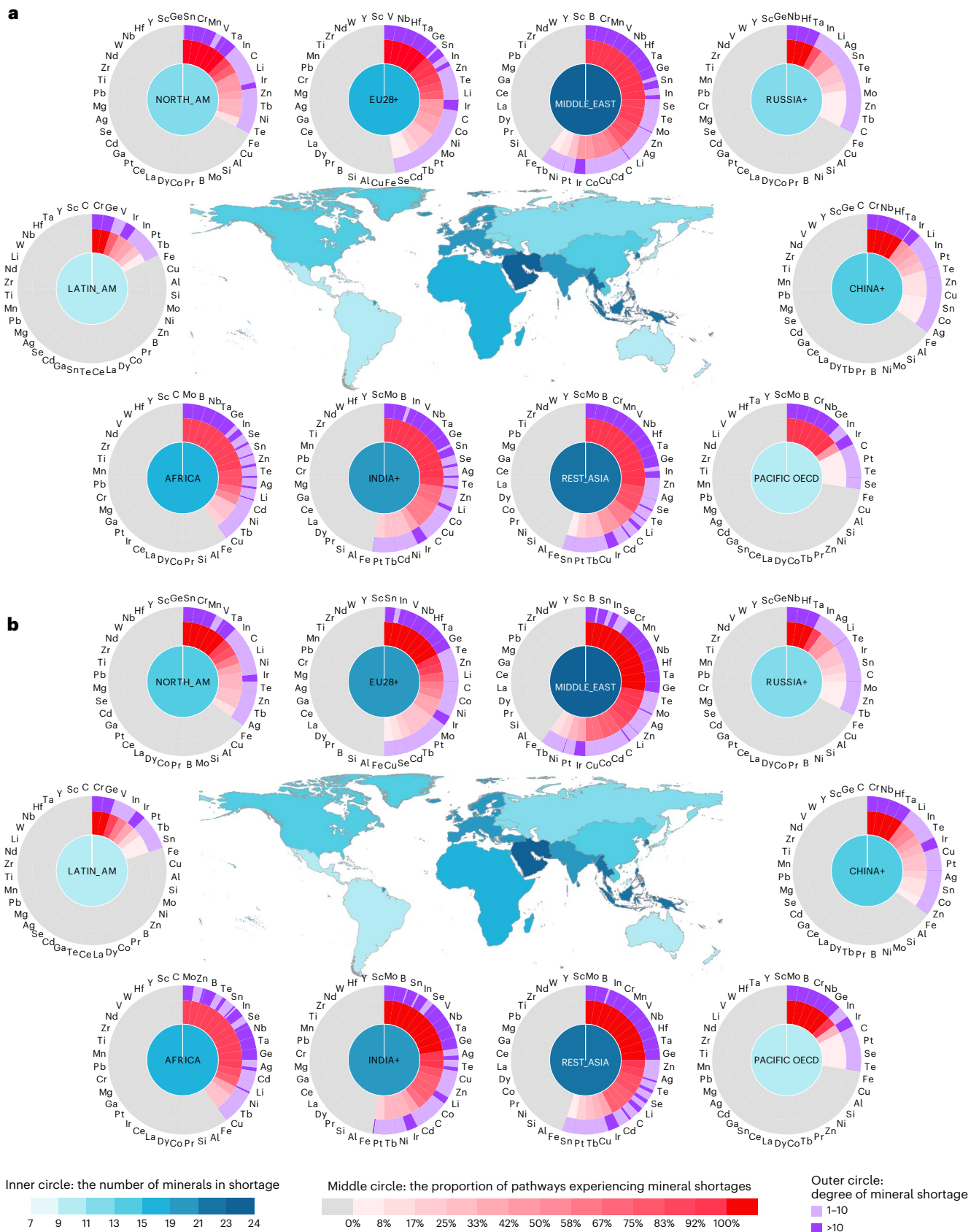


Fig. 3 | The types and degree of mineral shortages across the 10 IPCC-defined regions. a. The situation under the 2°C target. **b.** The situation under the 1.5°C target. The result presented here corresponds to the moderate scenario. For results under aggressive and conservative scenarios, refer to Supplementary Figs. 8 and 9. The inner circle represents the number of minerals in shortages

for a particular region. The middle circle represents the proportion of pathways experiencing mineral shortages. The outer circle refers to the degree of mineral shortage, which is measured by the times of mineral demand to available mineral reserves (geographic). Details regarding IPCC-defined regions are provided in Supplementary Table 1.

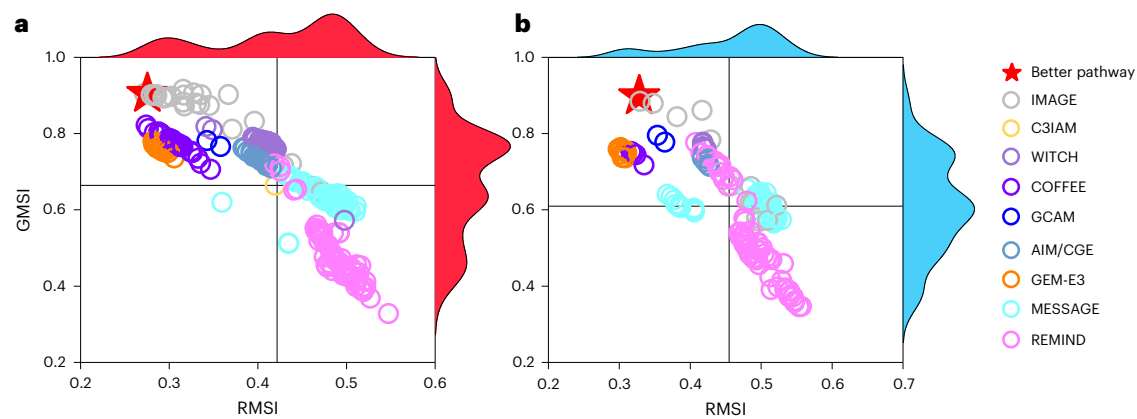


Fig. 4 | Performance of the 557 mitigation pathways in coordinating mineral resource security with climate targets. a, The performance of 367 pathways for the 2 °C target with respect to the GMSI and RMSI. **b,** The performance of 190 pathways for the 1.5 °C target with respect to the GMSI and RMSI. The result presented here corresponds to the moderate scenario. For results under aggressive and conservative scenarios, refer to Supplementary Figs. 10 and 11. The star represents the two better pathways identified in this study. Among them, IMAGE 3.2_SSP2_SPA2_19I_LI is the pathway that would limit warming to

1.5 °C with no or limited overshoot (>50%), while IMAGE 3.0_EN_NPi2020_1400 is the pathway that would limit warming to 2 °C (>50%). The horizontal and vertical lines represent the average values of GMSI and RMSI, respectively, calculated across all assessed pathways. The histograms positioned along the top and right edges of each panel depict the frequency distributions of pathways according to their GMSI and RMSI scores. A higher GMSI score along with a lower RMSI value indicates that the pathway achieves a better balance between resource and climate security at both global and regional levels.

encompassing accelerated technological innovation, enhanced recycling systems and regional trade optimization.

Optimal emissions reduction strategies

On the basis of the integrated performance of global risks and regional disparities, two optimal pathways that can better balance emissions reduction, resource constraints and regional equity are identified: IMAGE 3.2_SSP2_SPA2_19I_LI for 1.5 °C and IMAGE 3.0_EN_NPi2020_1400 for 2 °C (Fig. 4 and Supplementary Fig. 12). The socio-economic and technological portfolios of these two pathways are illustrated in Supplementary Figs. 13 and 14^{25,26}. Their development paradigms suggest that timely and reasonably paced emission reductions are key to minimizing mineral shortage risks. Under the 2 °C pathway, global CO₂ emissions are expected to decline by an average of 0.81 Gt annually from 2020 to 2050, then by 0.36 Gt per year until 2100. For energy technologies, renewable energy sources (wind, solar, hydro and biomass) would reach 65% of global installed capacity by 2050, with wind and solar power comprising approximately 62% of this total. Meanwhile, the share of global coal power capacity is expected to decline at a relatively gradual rate, averaging -0.5% annually until 2050, and the fossil fuel plants equipped with carbon capture and storage (CCS) need to contribute around 6.6%. By 2050, global energy storage installations are anticipated to reach nearly 380 GW. This technology portfolio suggests that overreliance on solar and wind energy is not an ideal solution owing to mineral constraints. Instead, fostering a diversified low-carbon technology portfolio is crucial. In addition, keeping the annual growth rate of the electricity demand at around 1.32% from 2020 to 2100 is also the key, indicating that a simultaneous and coordinated transformation of both low-carbon lifestyle changes and energy supply is imperative.

Mineral-focus strategies on the global scale

In the optimal pathways, cadmium and indium face shortages under the 1.5 °C target, while indium is in shortage under the 2 °C target. Addressing these shortages should synergize improvements in recycling technologies and mineral intensity reduction. However, achieving this would require unprecedented efforts. Specifically, the cadmium consumption intensity in nuclear power and PV technologies must sustain annual reductions of 0.49% and 2.49% relative to current levels, respectively, over four decades. Simultaneously, the composite cadmium recycling rate (considering both product recycling efficiency

and recovery of minerals) must maintain an annual increase of 0.1% for eight decades, reaching 19% by 2100. For indium, achieving the 2 °C target requires annual intensity reductions of 0.89% in nuclear power and 4.77% in solar PV through 2030, coupled with an 80-year continuous recycling rate increase of 0.92% by 2100. More aggressive technological advancements are necessary under the 1.5 °C scenarios (Supplementary Table 2).

Mineral-focus strategies regarding regional collaboration

Regional shortages are more severe than those on the global level, demonstrating that enhanced international cooperation is essential (Fig. 5). Considering the projected future demand, growth rates, coverage regions and severity of shortages, we optimize trade patterns for four critical minerals including nickel, copper, lithium and tin. Our model reveals that under the optimal 2 °C pathway, cumulative inter-regional trade volumes through 2100 are projected to reach 2.29 Mt for nickel, 39.54 Mt for copper and 0.76 Mt each for lithium and tin to address regional supply–demand imbalances. Trade structures will adapt to shifting demand patterns. On the export side, dominant regions may change. For example, RUSSIA+ leads nickel exports by mid-century, contributing 34% of global exports, but its export share is expected to decrease substantially by 2100, with NORTH_AM and LATIN_AM gaining larger shares (Fig. 5a). Lithium trade dynamics show even more pronounced shifts. By mid-century, regions are largely self-sufficient, with NORTH_AM as the top exporter (40% of global exports). By 2100, trade becomes more complex, with North America remaining a key exporter while RUSSIA+ and LATIN_AM also emerge as important exporters. On the import side, INDIA+ and AFRICA are central hubs for these four minerals, accounting for 26–62% and 14–28% of global accumulative imports, respectively. To meet mineral demand under the 1.5 °C pathway, even closer inter-regional collaboration is required.

Discussion

This study advances beyond fragmented analyses with a systematic evaluation of critical mineral constraints in global and regional energy transitions under the 1.5 °C and 2 °C targets, and proposes a dual-tier solution that can better balance emissions reduction, resource limitations and regional equity. Mineral shortage risks for 40 minerals of 17 energy technologies across all IPCC AR6 pathways are comprehensively investigated.

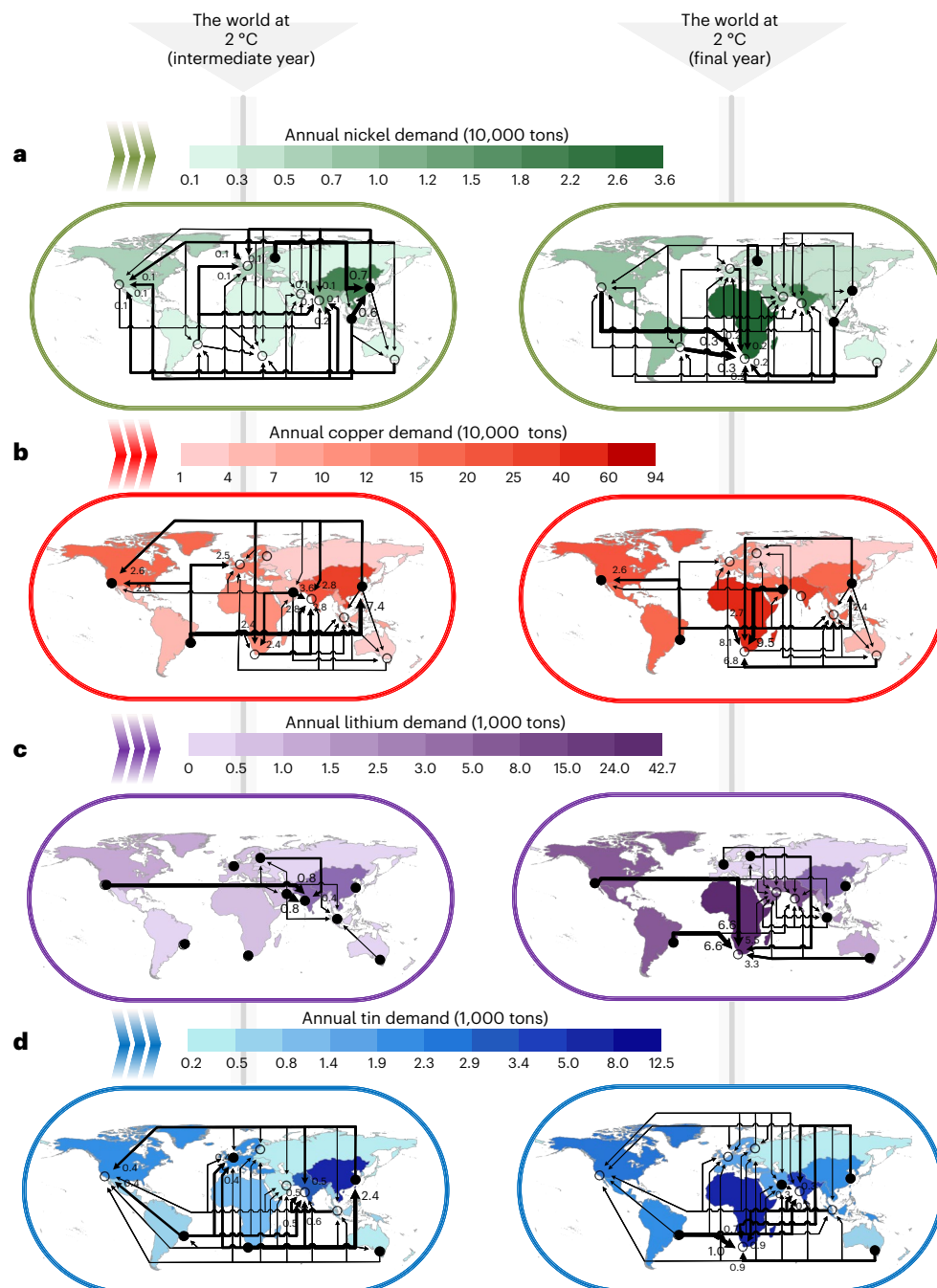


Fig. 5 | Regional mineral trade flows addressing shortages under identified optimal mitigation pathways. a–d. Results for nickel (a), copper (b), lithium (c) and tin (d) for 2 °C (left) and 1.5 °C (right). Considering the soaring demand, high growth rates, pivotal role in advancing key low-carbon technologies, critical minerals targeted by major economies and the diversity of reserve distribution, four major minerals—copper, nickel, tin and lithium—have been selected for the regional trade analysis. Furthermore, the study considers the potential increase

in mineral extraction costs (Methods). For all minerals, the intermediate year corresponds to 2050. Owing to earlier shortfall in global reserves, the final year for tin is indicated as 2099 for the 1.5 °C scenario. For the other three minerals, the final year is 2100. Regions without shading indicate a complete dependence on imports to meet their mineral demands, while regions with a black circle indicate partial self-sufficiency in mineral supply.

The results suggest that substantial mineral shortages may emerge, with 1–12 minerals potentially facing global deficits by 2100 under the moderate scenario (an average of 5), affecting multiple technologies including solar PV, energy storage batteries, wind and nuclear power. Critical gaps are observed for indium, cadmium and tin, where demand would exceed known reserves by 1.3–11.1 times. Tin shortages alone may limit the deployment of up to 32% of the anticipated global solar PV capacity under 2 °C pathways. Regional disparities further compound these challenges, with some developing regions

(for example, the Middle East, Africa and South Asia) facing potential shortages of up to 24 minerals. Nonetheless, mineral supply constraints may in turn reshape the trajectory of energy technology development, altering both the types and volumes of minerals required.

The identified shortages could be partially mitigated by technological substitution, for instance, replacing thin-film PV with highly efficient crystalline silicon to reduce reliance on indium, cadmium and tellurium, or developing cobalt-free batteries to ease dependence on nickel and cobalt. However, these shifts may exacerbate shortages of

tin, copper and lithium, presenting new risks in low-carbon technology development. Therefore, advancing alternative technologies and materials is essential to easing such constraints. Some efforts include the development of tin-free solders, sodium-based battery technologies, the use of electroplated copper and aluminium as substitutes for silver in solar cell metallization and various non-metallic or synthetic materials^{13,14,27}. These solutions may help reduce mineral demand but could also introduce secondary trade-offs, such as efficiency penalties, higher manufacturing costs and environmental concerns, which warrant continued innovation and further scrutiny.

The above-identified resource risks underscore the importance of providing a better emission reduction strategy from a mineral security perspective. Hence, this study navigates emission reduction pathways with minimum mineral constraints. The results show that, rather than over-relying on solar and wind, a diversified energy portfolio strategy is preferable, emphasizing the synchronized expansion of biomass, CCS and negative emissions technologies. Consequently, it is imperative to strengthen policy support and increase research and development investment in these technologies to accelerate innovation and applications. Beyond restructuring the energy mix, raising public awareness to promote low-carbon lifestyle changes is equally crucial for reducing the energy demand. To ensure this transition aligns with economic stability, maintaining an annual gross domestic product growth rate of approximately 2.7% by 2050 is recommended.

To fully mitigate the remaining global mineral resource risks, society must pursue transformative innovations beyond conventional mitigation efforts. Only when the indium intensity in nuclear and solar is reduced by at least 20% and 70%, and recycling efficiency is increased to at least 82% can the indium shortage be fully addressed to achieve the 2 °C target. Realizing these advancements requires systemic changes, such as immediate technological breakthroughs, implementation of producer responsibility mechanisms in energy systems and digital mineral labelling to enhance traceability and enable dynamic resource management. It is also essential to enhance exploration and innovation in underexplored regions such as Africa and Central Asia.

Close trade cooperation is also key for solving the intensified regional risks and ensuring climate goals and equity. Trade alliances and restrictions would further hinder developing countries' access to affordable minerals essential for their low-carbon transition, thereby exacerbating global inequities in climate action²⁸. For example, the AFRICA region is projected to face great shortages of critical minerals, especially lithium, nickel and tin. Achieving its regional climate targets depends crucially on importing these minerals from the NORTH_AM, LATIN_AM, MIDDLE_EAST and RUSSIA+ regions, particularly with over 83% of the lithium demand requiring imports from NORTH_AM, MIDDLE_EAST and RUSSIA+. A failure in international trade may prevent AFRICA from unleashing its abundant wind and solar potential, threatening global climate targets. To address these interconnected challenges, global economies must prioritize mutually beneficial cooperation; and developed nations, in addition to focusing on their own emission reduction, should assist developing nations, such as INDIA+, AFRICA+ and the MIDDLE_EAST, which started CO₂ reduction strategies later and will suffer graver resource crises and higher costs (Fig. 5).

All in all, the mineral shortages identified in this study highlight the importance of a diversified approach to low-carbon technologies, balanced socio-economic development, broader international cooperation on mineral trade and urgent technological innovation to ensure a successful low-carbon transition. However, there are still some limitations that deserve future research. This study focusses primarily on energy supply systems—the most mineral-intensive sector—meaning that our estimates represent a lower bound, as additional demand from sectors, such as electric vehicles, smart infrastructure and climate adaptation, could further exacerbate mineral shortages. For instance, increasing cooling needs for air conditioners in the Global

South (Africa, Southeast Asia and Latin America) under the 2 °C target could potentially widen shortages for aluminium, copper and iron by 2.54–26.59% (estimated following the methods in literature^{29,30}). Additionally, mineral production itself generates CO₂, with lifecycle emissions from the 40 studied minerals contributing -0.57% of the total emissions in the optimal 2 °C pathway (estimated following the literature¹⁸), suggesting the need for more in-depth analysis on this issue. Moreover, our analysis is subject to limitations arising from methodological assumptions. Specifically, it is grounded in current technological development trends and thus cannot fully capture potential shifts driven by resource scarcity, price fluctuations or technological breakthroughs (for example, alternative materials or emerging technologies). Given the uncertainty surrounding future innovation in a long-term circle, accurately characterizing its impact of such innovation remains a substantial challenge and a key direction for methodological improvement. Despite this, we have tried to capture the impact of technology dynamics and a clear 'seesaw effect' in technology mixes is identified (see sensitivity analysis in Supplementary Figs. 18 and 19), highlighting the complexity of low-carbon transitions and the need for coordinated planning—particularly in managing secondary risks from substitute technologies. Moving forwards, it is also important to incorporate emerging technologies, new mineral discoveries and more differentiated system classifications (for example, rooftop PV) to further update the findings.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41558-025-02373-3>.

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Methods

To identify the better pathways for achieving both climate and mineral resource security, a Global Resource Evaluation of Abatement Technologies model was developed (Supplementary Fig. 1). Initially, we accumulated 557 mitigation pathways to the 2 °C or 1.5 °C target from the IPCC AR6 database. For each pathway, we evaluated global and regional requirements for 40 critical minerals as used by 17 primary energy supply technologies (Supplementary Fig. 2) based on current technological development trends. Through a comparison of these requirements with the reserves of the chosen minerals, we anticipated the resource shortages and potential crises, that the world and its regions may face in a low-carbon future. To assess which mitigation pathways can concurrently balance both resource and climate security, we developed two indexes, namely the GMSI and the RMSI. On this basis, two better pathways are identified, but they are still facing potential shortage challenges that may remain. Consequently, we give trade-based solutions by developing a highly cooperative global trade optimization model. Furthermore, quantitative strategies for resource recovery and reducing mineral usage intensity are also provided.

IPCC pathways and scenario settings assessed in this study

The 557 IPCC AR6 pathways used in this study. To evaluate future mineral demand and potential shortages driven by energy technology deployment in the AR6 ensemble, we analyse mitigation pathways classified under the IPCC AR6 scenario database: C1 (below 1.5 °C with no or limited overshoot), C2 (below 1.5 °C with high overshoot), C3 (probably below 2 °C) and C4 (below 2 °C)³¹. These categories are further grouped into 1.5 °C (C1 and C2) and 2 °C (C3 and C4) pathways.

Before conducting our analysis, we conducted a systematic review of 17 key energy supply technologies in these pathways, excluding those with insufficient data for critical future power generation technologies—specifically PV, wind, hydropower and nuclear technologies, resulting in a final sample of 557 pathways—190 for 1.5 °C and 367 for 2 °C (Supplementary Fig. 1 and Supplementary Data 1–3).

Scenarios setting for mineral intensity and recycling. To represent the uncertainties of mineral intensity declines and recycling improvement, 80 scenarios are developed in this study by combing different levels of recycling efficiency and mineral intensity. For ease of understanding, we select three representative scenarios (Fig. 1a, red boxes) to be analysed in detail. All the results in the main text if not specifically mentioned are referring to that of the moderate scenario (see Supplementary Figs. 6–11 for aggressive and conservative scenario outcomes).

For intensity reduction, this study sets eight levels (R1–R8), indicating an initial linear decline from the baseline value to the most advanced level reported in the literature by a specified year, followed by a continued decline at half the previous rate until 2100. R1–R8 represent reaching the most advanced level in 2030, 2040, 2050, 2060, 2070, 2080, 2090 and 2100, respectively. R1 represents the case with the most rapid technology innovation and R8 is the case with the slowest innovation.

To establish these levels, we collected over 850 mineral intensity data points across 17 assessed technologies and 40 key minerals from 167 recent publications, including industry reports, media articles, expert consultations and government data. Baseline and advanced values were determined by first removing extreme data points (outliers beyond the 99th percentile), then using the average mineral intensity as the baseline and the minimum reported values as the advanced level^{32–42} (for more details, see Supplementary Data 5). Supplementary Figs. 15–17 illustrate the quantitative mineral usage per technology and how the selected baseline values compare with the literature-reported ranges.

For mineral recycling, the annual cumulative amount of minerals that can be recycled from retired technologies was calculated on the basis of the current composite recycling rate (considering product

recycling efficiency and recovery of minerals) for each technology. It is assumed that these recycled minerals can be efficiently reused in the construction of future technologies. Since baseline recycling rates vary across technologies, we set ten levels for the recycling rate increase by 2100, with a linear progression relative to baseline values, ranging from 0.05 to 0.5. A rate of <0.20 represents slow improvements, 0.20–0.40 indicates medium improvements and >0.40 suggests rapid improvements. The baseline values are sourced from established literature (Supplementary Data 6).

Evaluating the demand for minerals and their shortages on 557 mitigation pathways

The manufacture of the 17 assessed technologies relies on 40 critical minerals^{43,44}. This study systematically assesses global and regional demand and shortage risks of these 40 critical minerals across 557 IPCC AR6 pathways.

Mineral demand. We designated 2005, the earliest year documented in the IPCC AR6 database, as the baseline year and considered the installed capacity of all technologies in that year as the new installed capacity. Owing to the comparatively low global installed capacity of mineral-intensive technologies in 2005, the simplified assumption regarding the installed capacity in that particular year has a negligible effect on the outcomes of the model. To fill in the gaps in the scenario data, we linearly interpolated the installed capacity of each technology type for each year.

To ensure consistency with the original model assumptions for each IPCC pathway, we created a 557×17 heterogeneous lifetime matrix to reflect the diversity of technology lifetimes across all IPCC AR6 pathways (Supplementary Data 4). We extracted lifetime data for 17 technologies from 557 pathways in the IPCC AR6 database. For missing data, we first used lifetime assumptions from other pathways within the same integrated assessment model family, and if unavailable, applied the most frequently reported value across all pathways. Furthermore, assuming that the life curves of different technologies conform to a Weibull distribution⁴⁸, we integrated the installed stock and retirement rate of each technology across various pathways to calculate the new installed capacity and retired capacity for each technology in each year from 2005 to 2100.

Recognizing the substantial heterogeneity among various technologies, we subsequently implemented several assumptions to transform integrated assessment model projections for PVs, lithium-ion battery and electrolysis for hydrogen production into a more detailed breakdown of specific technologies. PVs were divided into four distinct types: amorphous-silicon, crystalline-silicon, cadmium telluride and copper indium gallium diselenide; three lithium-ion battery types were identified, lithium iron phosphate battery, nickel cobalt manganese oxide battery and lithium cobalt oxide battery; and electrolysis for hydrogen production was divided into alkaline water electrolysis, proton exchange membrane water electrolysis and solid oxide water electrolysis^{45–47}. Their technical specifications and initial market shares are provided in Supplementary Data 11.

The annual mineral demand for a specific technology was determined by multiplying its new installed capacity in a given year by its corresponding mineral consumption intensity. Similarly, the mineral recovery amount was calculated by multiplying the mineral content of retired equipment by the corresponding composite recycling rate. Notably, the mineral content should reflect the consumption intensity at the time of deployment, not retirement. This study does not emphasize dynamic price impacts on mineral demand, as the demand for low-carbon transitions remains inherently rigid, showing limited sensitivity to short-term price fluctuations, as evidenced by historical demand–price elasticity⁴⁸.

Specifically, for a particular year t , the demand for the j th mineral by the i th technology in the r th region under the n th pathway ($D_{n,r,t}^j$)

was determined by matching technology demand with the corresponding mineral consumption, following equation (1)

$$D_{A_{n,r,t}}^j = \sum_i^{N_i} (C_{i,t}^j \times P_{A_{n,r,t}}^j) \quad (1)$$

where $C_{i,t}^j$ denotes the unit consumption of the j th mineral resource by the i th technology in year t , which is measured in t GW⁻¹, t GW⁻¹ or t MtCO₂⁻¹, and $P_{A_{n,r,t}}^j$ is the new installed capacity of the i th technology in the r th region under the n th pathway in year t , which is measured in gigawatts, gigawatt hours or million tonnes of CO₂. N_i is the number of technologies.

For a particular year t , the amount of recycled mineral for the j th mineral by the i th technology in the r th region under the n th pathway ($D_{R_{n,r,t}}^j$) was determined by the composite recycling rate and the amount of mineral contained in the retired installed capacity, as follows:

$$D_{R_{n,r,t}}^j = R_t^j \times \sum_i^{N_i} M_{R_{n,r,t}}^j \quad (2)$$

$$M_{R_{n,r,t}}^j = \sum_{t'=2005}^{t=t} (C_{i,t'}^j \times P_{R_{n,r,t'}}^j) \quad (3)$$

where R_t^j denotes the recycling rate of the j th mineral resource, recycling rates for each mineral increase linearly from 2020 to 2100, and using the moderate scenario as an example, we assume that recycling rates for all minerals increase by 0.25 in 2100 compared with 2020. Further, we calculate the recovery rate for each mineral in each year³³. $M_{R_{n,r,t}}^j$ is the total amount of the j th mineral contained in the retired equipment of the i th technology in the r th region under the n th pathway in year t , measured in tonnes. $C_{i,t'}^j$ denotes the unit consumption of the j th mineral resource by the i th technology in year t' , measured in t GW⁻¹, t GW⁻¹ or t MtCO₂⁻¹. $P_{R_{n,r,t'}}^j$ is the retirement capacity of the i th technology that began operating in year t' in the r th region under the n th pathway, measured in gigawatts, gigawatt hours or million tonnes of CO₂.

The global demand ($D_{A_{n,w,t}}^j$) for the j th mineral required for the n th pathway in year t is calculated by summing up regional demands, following equation (4)

$$D_{A_{n,w,t}}^j = \sum_r^{N_r} D_{A_{n,r,t}}^j \quad (4)$$

where N_r represents the number of regions.

Similarly, the global amount of recycled mineral ($D_{R_{n,w,t}}^j$) for the j th mineral resource under the n th pathway in year t is calculated by aggregating the regional amounts

$$D_{R_{n,w,t}}^j = \sum_r^{N_r} D_{R_{n,r,t}}^j \quad (5)$$

Using these equations, we estimated the annual demand and recovery for 40 minerals, at the regional and global levels, for each of the 557 emissions reduction pathways.

Amount of mineral available for use in the future. The projected maximum amount of the j th mineral available for future use by the energy systems of the r th region in year t ($M_{\text{useable},r,t}^j$) is typically equal to the total reserves of the j th mineral in the same region ($S_{r,t}^j$) (equation (6)). Given the widespread utilization of minerals in various global economic activities, the global maximum amount of mineral available for use in year t ($M_{\text{useable},w,t}^j$) can be computed by multiplying the total reserves of the j th mineral worldwide ($S_{w,t}^j$) by the maximum proportion of the

j th mineral available for energy systems in the same year ($K_{\text{energy},t}^j$) (equation (7)).

$$M_{\text{useable},r,t}^j = S_{r,t}^j \quad (6)$$

$$M_{\text{useable},w,t}^j = S_{w,t}^j \times K_{\text{energy},t}^j \quad (7)$$

Global and national reserves were primarily taken from US Geological Survey data for 2021⁴⁹. For any missing data in our study, we obtained the data through a commercial data service provided by the China Bulk Commodity. Reserves for the ten regions were obtained by adding national level data. The proportion ($K_{\text{energy},t}^j$) is calculated using the historical average for each mineral, which was based on prior research conducted by Charpentier Poncelet et al.⁵⁰ (Supplementary Data 7). We aggregated the mineral usage proportions in different applications to obtain the proportion of mineral available for energy systems' consumption. More specifically, for specialty minerals, the usage proportion was predominantly determined by their application ratios in magnets, electronics and batteries⁵⁰. Certain uses of the minerals were not included in the analyses. Specifically, in terms of ferrous and non-ferrous minerals, the study does not consider their applications in end-use sectors such as transportation and household appliances. Similarly, for precious minerals, applications in the jewellery sector and other end-use sector areas are not considered.

Given the possibility of an accelerated energy transition, the proportion of mineral consumption by the energy system relative to the total societal mineral consumption may increase. Predicting changes in this ratio is challenging; therefore, it is assumed to remain constant in this study.

Degree of mineral shortages. Owing to the unpredictable and irregular nature of mineral discoveries, this study utilizes the 2021 proven economic reserves as a benchmark and concentrates on assessing whether the existing mineral reserves can adequately meet the escalating demand for minerals arising from the low-carbon transition in the future. The degree of mineral shortage (SR) is measured by the ratio of net mineral demand to available mineral reserves (equations (8) and (9)). The annual net demand for minerals is determined by subtracting the mineral recycling volume in a given year from the newly generated demand for minerals in that year^{13,51}. A ratio of more than 1 means that the amount of available mineral falls short of the amount required. This makes it more challenging to implement the assessed mitigation pathway and reduces the likelihood of meeting climate targets.

For the globe in year t

$$SR_{n,w,t}^j = \left(\sum_{t=2021}^{N_t} (D_{A_{n,w,t}}^j - D_{R_{n,w,t}}^j) \right) / M_{\text{useable},w,t=2021}^j \quad (8)$$

For each region in year t

$$SR_{n,r,t}^j = \left(\sum_{t=2021}^{N_t} (D_{A_{n,r,t}}^j - D_{R_{n,r,t}}^j) \right) / M_{\text{useable},r,t=2021}^j \quad (9)$$

Identifying the better pathways balancing emission reduction with mineral constraints from both a global and regional perspective

To navigate the optimal emissions reduction strategies with the minimum mineral risks across all IPCC AR6 pathways aligned with the 1.5 °C and 2 °C targets, we establish two indices: GMSI and RMSI. These indices systematically evaluate how deceleration rates, socio-economic changes and technological combinations impact mineral risks across pathways by considering mineral shortages, severity and timing at both global and regional levels.

GMSI. Given the variability in the severity and type of mineral shortage for each pathway, we adopted a multi-criteria decision-making approach on the basis of ideal target similarity⁵². This approach determines a so-called positive ideal solution and a negative ideal solution. The best solution is that which is nearest the positive ideal and furthest from the negative ideal. A GMSI was calculated for each path. This was initiated by creating normalized indicators of projected mineral demand (equation (10)).

$$x_n^j = (\max(D_n^j) - D_n^j) / (\max(D_n^j) - \min(D_n^j)) \quad (10)$$

where $\max(D_n^j)$ and $\min(D_n^j)$ represent the maximum and minimum values of j th mineral demand on the n th pathway, respectively.

Subsequently, the weighted decision matrix was computed (equation (11)), along with the positive and negative ideal solutions (equation (12)). If x_n^j represents the normalized values, a weighted decision matrix is calculated as follows:

$$z_n^j = w_j \times x_n^j \quad (11)$$

where z_n^j denotes weighted values in the normalized decision matrix, and w_j is the relative importance (weight) of the j th mineral.

$$z_j^+ = \max(z_n^j), z_j^- = \min(z_n^j) \quad (12)$$

where z_j^+ and z_j^- denote the positive and the negative ideal solutions for the j th mineral, respectively.

Then, the Euclidean distance between each solution and the ideal solution was determined as

$$d_n^+ = \sqrt{\sum_{j=1}^{N_j} (z_j^+ - z_n^j)^2}, d_n^- = \sqrt{\sum_{j=1}^{N_j} (z_n^j - z_j^-)^2} \quad (13)$$

where d_n^+ and d_n^- denote the distance of the n th pathway from the positive and negative ideal solutions, respectively, and w_j denotes the relative importance (weight) of the j th mineral.

Finally, the GMSI score of each pathway was calculated using equation (14).

$$\text{GMSI}_n = d_n^+ / (d_n^+ + d_n^-) \quad (14)$$

When assigning the weights to the minerals (that is, w_j), three methods were applied:

- (1) Equal weights for all 40 minerals.
- (2) Minerals were divided into two groups with equal internal weights. The group of minerals in shortages accounted for 60%, and the remaining minerals accounted for 40%.
- (3) As above, but the group of minerals in shortages accounted for 70%, and the remaining minerals accounted for 30%.

To obtain the final index value, the scores under the three weight combinations were averaged. A higher score indicates better performance (lower degree of mineral shortages).

RMSI. In addition to global-level challenges, regional shortages can also provoke crises. To characterize the situation arising for each mitigation pathway, we developed an RMSI based on the standardization of three indicators:

- (1) The proportion of minerals in shortage along the pathway
- (2) The average duration of shortages for each mineral
- (3) The maximum shortage degree for each mineral

The higher the value of RMSI, the greater the risk of mineral resource shortages in certain regions: either higher overall degree of mineral shortage, a longer duration or more intensified shortages of a specific mineral.

Mineral trade optimization model based on regional co-operation

The potential mineral resource shortages, identified in the better pathways, were addressed using our mineral trade optimization model on the basis of regional co-operation. This model considers the economic, social, and political interactions in geographical space, with the aim of minimizing global mineral trade costs. By analysing factors such as mineral demand, production capacity, extraction costs and transportation costs in each region, optimal trades (volume and path) are identified. We assumed that the major international transshipment ports, located within resource-rich countries across the regions, would serve as the primary interface for inter-regional mineral trades. In addition, the simulated patterns of mineral trade were limited by factors such as resource endowment, historical trade patterns and the need for trade diversification and geopolitical risks⁵³.

Objective function. The objective function of the model is to minimize the annual global cost of mineral consumption, for each mineral, considering both extraction and transportation costs (equation (15)). The annual demand of each mineral in a region can be met through a combination of international trade and domestic supply. In the case of international trade, the model considers the extraction costs and transportation costs associated with sourcing minerals from their respective supply locations (Supplementary Data 9 and 10). Conversely, for domestic self-sufficiency, only the extraction costs are considered.

$$\min_{t_1 \leq t \leq t_2} f = \sum_{j=1}^{10} \sum_{i=1}^{10} \{X_{ij}(t) \times [\text{pcost}_i + \text{tcost}_{i,j} \times \text{dis}_{i,j}]\} \quad (15)$$

where $X_{ij}(t)$, the decision variable in this model, represents the amount of mineral imported, in year t , by the i th region from the j th region ($i \neq j$) or produced domestically ($i = j$) to meet its own demand. pcost_i represents the unit cost of mineral extraction in the i th region, $\text{tcost}_{i,j}$ represents the unit transportation cost for shipping the corresponding mineral from the j th region to the i th region and $\text{dis}_{i,j}$ represents the shipping distance between these two regions (when $i = j$, $\text{dis}_{i,j} = 0$).

According to the Hotelling rule in resource economics, the prices of non-renewable resources like minerals are anticipated to rise as reserves diminish. However, accurately predicting costs and prices over the lengthy timeframe of 2020 to 2100 is extremely challenging. This study adopts the average rate of mineral price changes over a 70-year period from 1950 to 2020 as a representation of future trends in mineral costs. This average rate is then incorporated into the trade optimization model to address the challenges associated with the dynamic mineral production costs.

Constraint conditions. The optimal global mineral trade cooperation is constrained by factors such as regional resource endowments, the balance of global supply and demand, and historical international trade relations. To achieve a successful global low-carbon energy transition and meet the global deployment requirements of energy technologies, it is of utmost importance to ensure the total mineral supply equal to or greater than the demand for that mineral in that year, as outlined in equation (16).

$$\sum_{i=1}^{10} \text{Supply}_i(t) \geq \sum_{j=1}^{10} \text{Demand}_j(t) \quad (16)$$

where $\text{Supply}_i(t)$ represents the total amount of mineral that the i th region exports to other regions or supplies for local consumption in year t . $\text{Demand}_j(t)$ represents the annual demand for the same mineral in the j th region in that year.

For each region, the annual supply amount consists of both the minerals supplied to other regions and the minerals supplied domestically (equation (17)).

$$\text{Supply}_i(t) \geq \sum_{j=1}^{10} X_{i,j}(t) \quad (17)$$

For each year, the supply in a specific region cannot exceed the available reserves, as expressed by equation (18).

$$\text{Supply}_i(t) \leq \text{Reserve}_i(t) \quad (18)$$

where $\text{Reserve}_i(t)$ denotes the available reserves of the i th region in year t .

As minerals are non-renewable, the mineral reserves of each region need to be updated in each period. Specifically, the reserves of a mineral in a region at time t should be equal to those in the previous period minus the total amount of that mineral that the region supplied to other regions or consumed locally in the previous period. This can be expressed mathematically by equation (19).

$$\text{Reserve}_i(t+1) = \text{Reserve}_i(t) - \text{Supply}_i(t) \quad (19)$$

For each region, the net annual increase in mineral demand should be met through regional trade and domestically supply, as follows:

$$\text{Demand}_j(t) \leq \sum_{i=1}^{10} X_{i,j}(t) \quad (20)$$

Furthermore, trade sanctions, geopolitical crises and trade protection measures have the potential to impact trade dynamics between regions profoundly^{53,54}. In the present study, we predict future trade tendencies on the basis of previous trade partnerships between regions and the resource endowments of each region. Additionally, we introduce proportion constraints on inter-regional import and export trade, considering the trend of international trade diversification. These constraints aim to reduce the risks associated with overreliance on specific regions for mineral sourcing. More precisely, the maximum quantity of a specific mineral that region i can import from region j depends on a maximum degree of regional trade dependency between these two regions ($\text{rate}_{i,j} \in [0, 1]$), as articulated by equation (21).

$$X_{i,j}(t) \leq \text{Demand}_j(t) \times \text{rate}_{i,j} \quad (21)$$

Here, the maximum degree of dependency, denoted as the maximum import proportion of region i from region j ($\text{rate}_{i,j}$), is determined by considering various factors, including resource endowment, self-sufficiency, historical trade patterns and the need for trade diversification. The rules for setting this proportion are provided in Supplementary Tables 3–7.

In addition to the aforementioned constraints, the decision variables must satisfy non-negativity constraints. Another binding constraint is that global available reserves in each period cannot fall below zero. If all resources of a mineral are exhausted globally, a regional resource crisis cannot be alleviated through trade. Alternative solutions, such as improved resource recovery, reduced mineral consumption intensity, the recycling of mine tailings and the adoption of disruptive technology, would need to be explored.

Data availability

The IPCC AR6 database used in this study is available via IIASA at <https://data.ene.iiasa.ac.at/ar6/>. We extracted comparable global and regional pathways data from the IPCC AR6 database³¹. The baseline and advanced values for mineral intensity and recycling of each technology were sourced from peer-reviewed publications (Supplementary Data 5). The reserve data for 40 minerals were obtained from the USGS (<https://www.usgs.gov/>) and China Bulk Commodity (<https://en.cbci.com/>) (Supplementary Data 8). The historical average proportion of

minerals consumed by energy systems is based on ref. 50 (Supplementary Data 7). Mineral commodity trade data are available via the UN Comtrade Database at <https://comtradeplus.un.org/>. The base maps in this study are sourced from the Resource and Environmental Science Data Platform of the Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences. The Supplementary Data supporting this study are available via Zenodo at <https://doi.org/10.5281/zenodo.15482842> (ref. 55).

Code availability

The matrix manipulation was used to solve our model with the equations illustrated in the Methods section of our manuscript. The code is written and calculated by Python 3.9 and PyCharm Community Edition 2022.3.2 and related description of matrix manipulation is available via Python at <https://docs.python.org/3/library/index.html>. The linear programming solution procedure was used to solve our mineral trade optimization model with the equations illustrated in the Methods section of our manuscript. The chronological linear optimization solver is based on Python 3.6 and Gurobi Optimizer (Gurobi 9.0.0). The code and related description of Gurobi Optimizer are available via Gurobi at <https://www.gurobi.com/documentation/current/refman/index.html>. The code supporting this study is available via Zenodo at <https://doi.org/10.5281/zenodo.15482842> (ref. 55).

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Author contributions

Y.-M.W., L.-C.L., J.-N.K. and B.Y. conceived the study. J.-N.K., Y.Z., S.P., S.X., H.Y., Y.J., X.T., W.C., J.L., H.C., M.D., X.L., F.Y. and J.Q. contributed to the data collection and processing. Y.Z., S.P., X.Q., L.Z., D.W., L.-C.L., J.-N.K., Q.-M.L. and H.L. designed and performed the models’ runs. Y.Z., S.P., H.Y., S.X., L.-C.L., J.-N.K. and B.Y. implemented the data presentation and visualization. Y.-M.W., L.-C.L., J.-N.K., Y.Z., S.P., S.X., Y.J., H.Y., X.Q., X.T., Q.-M.L., H.L. and B.Y. contributed to the interpretation of the results. Y.-M.W., L.-C.L., J.-N.K., Y.Z., S.P., S.X., Y.J., H.Y., X.Q. and X.T. prepared the first draft. Y.-M.W., L.-C.L., J.-N.K., Y.Z., S.P., H.Y. and B.Y. contributed to the Supplementary Material. Y.-M.W., L.-C.L., J.-N.K., H.L. and B.Y. worked on the review and editing. All authors approved and contributed to writing the paper.

Competing interests

The authors declare no competing interests.

Additional information

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